

Software Engineering

Lab 1: Requirements Engineering & UML Use-Case Modelling

PROBLEM STATEMENT #56 | Media, Events & Community

Peer Skill Exchange & Mentorship Network

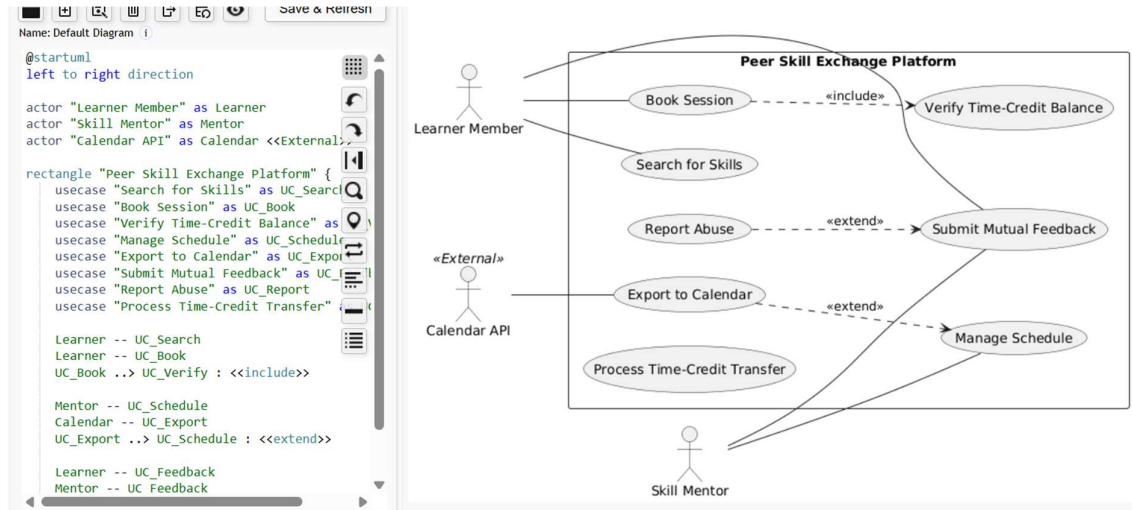
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1.Requirements Table

	A	B	C	D	E	F
1	Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
2	FR-001	Functional	The system shall maintain a time-credit balance for each member, crediting 1 hour upon verified completion of a teaching session.	High	Pass: Time credit transfers from learner to mentor after session sign-off. Fail: Member with 0 credits books session.	Core time-banking currency mechanism.
3	FR-002	Functional	The system shall allow members to search for mentors using filters for skill category, availability, and average feedback rating.	High	Pass: Searching "Python" returns only mentors offering Python with their respective ratings and open slots.	Ensures learners can find relevant skills effectively.
4	FR-003	Functional	The system shall temporarily place a hold on a Learner's time-credit when they request to book a session.	High	Pass: Learner's available balance decreases by 1 immediately upon booking request.	Prevents double-spending of limited time-credits before a session is completed.
5	FR-004	Functional	The system shall mandate both the Learner and Skill Mentor to submit a mutual feedback rating before the time-credit transfer is finalized.	Medium	Pass: Credit remains pending until both parties submit a 1-5 star rating. Fail: Credit transfers with only one party's feedback.	Validates session completion and maintains platform quality.
6	FR-005	Functional	The system shall provide a dashboard displaying upcoming sessions, past session history, and the current time-credit balance.	Medium	Pass: Dashboard accurately reflects real-time credit balance and chronologically lists scheduled sessions.	Provides users with necessary account oversight.
7	NFR-001	Nonfunctional	The session scheduling calendar must sync seamlessly with Google Calendar and Outlook via iCal export feeds.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Facilitates schedule management outside the platform.
8	NFR-002	Nonfunctional	The platform's search function shall return filtered mentor results in under 2 seconds during peak load (10,000 concurrent users).	Medium	Pass: Load testing logs show 95th percentile response time is ≤ 1.9 seconds.	Ensures quick service and prevents user drop-off during peak traffic.

2. UML Use-case diagram



3. Use-Case Flow Specification

Use Case: Book a Skill Session Primary Actor: Learner Member Secondary Actor: Skill Mentor

Preconditions:

- The Learner Member is authenticated and logged into the system.
- The Learner Member has a time-credit balance greater than 0.
- The Skill Mentor has open availability slots on their calendar.

Postconditions:

- A session is scheduled on both the Learner's and Mentor's platform calendars.
- One time-credit is temporarily held/deducted from the Learner's available balance.
- A notification is sent to the Skill Mentor.

Main Success Scenario:

1. The Learner navigates to a specific Skill Mentor's profile.
2. The Learner selects an available 1-hour time slot from the Mentor's calendar and clicks "Book Session".
3. The system invokes the «include» Verify Time-Credit Balance use case to ensure the Learner has at least 1 credit.
4. The system validates the credit and temporarily places a hold on 1 time-credit from the Learner's account.

5. The system updates the calendar to mark the slot as "Booked".
6. The system sends a booking confirmation notification to both the Learner and the Skill Mentor.
7. The system displays a success message to the Learner.

Alternate Flow : Insufficient Time-Credits (Fails at Step 3)

- 3a. The system detects the Learner's time-credit balance is 0.
- 3b. The system halts the booking process.
- 3c. The system displays an error message: "Insufficient time-credits. Please teach a session to earn credits."
- 3d. The use case terminates without scheduling the session.